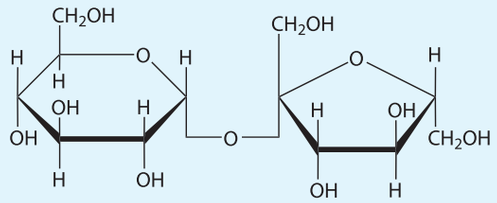
14.1.5 Disaccharides

Disaccharides are sugars that yield two molecules of monosaccharides on hydrolysis. This reaction is usually catalysed by dilute acid or enzyme. Disaccharides have general formula $C_n(H_2O)_{n-1}$.In disaccharides two monosaccharides are linked by *oxide linkage* called '*glycosidic linkage*', which is formed by the reaction of the anomeric carbon of one monosaccharide reacts with a hydroxyl group of another monosaccharide.

Example: Sucrose, Lactose, Maltose

Sucrose: Sucrose, commonly known as table sugar is the most abundant disaccharide. It is obtained mainly from the juice of sugar cane and sugar beets. Insects such as honey bees have the enzyme called invertases that catalyzes the hydrolysis of sucrose to a glucose and fructose mixture. Honey in fact, is primarily a mixture of glucose, fructose and sucrose.



Sucrose

(α-D-glucopyranosyl-β-D-fructofuranoside)

Figure 14.9 Structure of sucrose

On hydrolysis sucrose yields equal amount of glucose and fructose units.

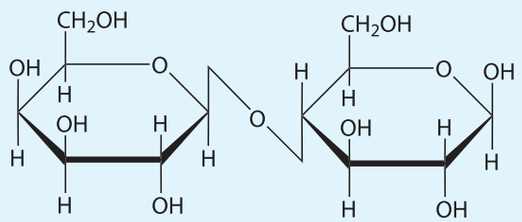
$$\text{Sucrose} \xrightarrow{\text{Invertase}} \text{Glucose+Fructose}$$

Sucrose (+66.6°) and glucose (+52.5°) are dextrorotatory compounds while fructose is levo rotatory (-92.4°). During hydrolysis of sucrose the optical rotation of the reaction mixture changes from dextro to levo. Hence, sucrose is also called as invert sugar.

Structure:

In sucrose, C1 of α-D-glucose is joined to C2 of β-D-fructose. The glycosidic bond thus formed is called α-1,2 glycosidic bond. Since, both the carbonyl carbons (reducing groups) are involved in the glycosidic bonding, sucrose is a non-reducing sugar.

Lactose: Lactose is a disaccharide found in milk of mammals and hence it is referred to as milk sugar. On hydrolysis, it yields galactose and glucose. Here, the β-D–galactose and β-D–glucose are linked by β-1,4 glycosidic bond as shown in the figure 14.10 . The aldehyde carbon is not involved in the glycosidic bond hence, it retains its reducing property and is called a reducing sugar.



Lactose

(β-D-galactopyranosyl-(1→4)β-D-glucopyranose)

Figure 14.10 Structure of Lactose

XII U14-Biomolecules_New.indd 247 11-01-2021 16:37:46